

SIH26060 — Simple Guide

Digital Platform for Remote Management of Indian Antarctic Research Stations

Ministry: Earth Sciences (MoES) | **Track:** Software | **Prize:** ₹1,00,000

1. What is the Problem? (In Simple Words)

India runs research stations in Antarctica (like Maitri and Bharati). These stations are extremely remote — no easy internet, no nearby city, no quick way to send supplies.

Right now, managing these stations (food stock, equipment, staff schedules, communication with India HQ) is likely done manually or through scattered tools (spreadsheets, emails, radio calls).

The core problem: There is no single, simple digital system to help station staff and India-based coordinators manage day-to-day operations remotely and efficiently.

2. What is the Solution Expected?

A **web-based management platform** that acts like a "control room dashboard" for an Antarctic station. It should let staff and coordinators:

Feature	What it does
 Inventory Tracker	Track food, fuel, medical supplies, equipment — know what's running low
 Staff & Duty Roster	Manage who's doing what shift, health status, rotation schedule
 Resupply Planner	Predict when the station will run out of key supplies, so ships/flights can be planned in advance
 Communication Log	Keep a record of messages, reports, and updates sent to/from India HQ
 Alerts & Anomaly Detection	Warn if supplies are depleting faster than normal, or equipment usage looks abnormal
 Dashboard	One clean screen showing station health at a glance

Where AI/ML comes in (your strength): Use simple prediction models to:

Forecast when supplies will run out (based on usage rate)

Detect unusual patterns (e.g., fuel used too fast → possible leak/fault)

This keeps AI *useful*, not just decoration.

3. Why This is a Good Pick for You

✅ **No hardware needed** — it's a pure software/web dashboard ✅ **Few teams choose this** — most people rush to Health/Education PSs, so less competition ✅ **Matches your skills** — full-stack (React/Node or similar) + basic ML (forecasting, anomaly detection) ✅ **Easy 90-second demo** — "This is a dashboard that helps Antarctic station staff manage supplies and predict shortages before they happen." ✅ **Strong resume line** — "Built a logistics & resource management platform for India's Antarctic research program"

4. How to Build It (Simple Plan)

Step 1: Define Scope (Day 1)

Don't try to build everything. Pick 3 core modules to build really well:

Inventory tracking + low-stock alerts

Staff duty roster

AI-based supply forecasting (simple regression/time-series model)

Step 2: Tech Stack (Keep it Simple)

Frontend: React.js (dashboard, charts using Chart.js/Recharts)

Backend: Node.js + Express (or Django if your team prefers Python)

Database: MongoDB or PostgreSQL

AI/ML: Python (scikit-learn) for forecasting — can run as a separate microservice

Hosting for Demo: Any free-tier cloud (Render, Vercel, Railway) or just run locally for the pitch

Step 3: Build Order

Set up database schema (Stations → Inventory, Staff, Logs)

Build backend APIs (CRUD for inventory, staff, logs)

Build frontend dashboard (tables + charts)

Add ML model: feed historical usage data → predict days-until-shortage

Add anomaly detection: flag unusual consumption spikes

Polish UI, add sample "dummy station data" for the demo

Step 4: Prepare the Pitch

1 slide: The Problem (remote station = poor visibility into supplies/staff)

1 slide: Your Solution (dashboard + AI prediction)

Live Demo: Show inventory dropping → system predicts shortage → alert triggers

1 slide: Impact (faster resupply planning, safer station operations, cost savings)

5. Important Next Step (Do This Before Finalizing)

⚠️ This guide is based on the **PS title only** from a curated list. Before you commit your team:

Go to the **official portal: sih.gov.in**

Search for **SIH26060** and read the *full official problem statement* (background, detailed description, expected solution)

Confirm if they mention any **dataset they will provide** — this changes your build plan

Confirm your **6-member team** (with 1 mandatory female member) is finalized before registration

Once you confirm the official PS text matches this scope, I can help you build the actual pitch deck, system architecture diagram, or even scaffold the starter code.